TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders
Linear combinations: make one variable's coefficients opposites, then add the equations.
Multiply every term, including the number on the right side.
Fractions: multiply by the LCD first. Variable alone? Substitution is faster.

1. Solve: $x + y = 20$ and $\frac{1}{2}x + \frac{1}{4}y = 7$.

Tip: Multiply by 4 to clear the fractions.

A. (8, 12)

B. (12, 8)

C. (10, 10)

2. Which method is most efficient for $4x - 3y = 5$ and $2x + 3y = 7$?

Tip: Look for opposite coefficients or a variable alone.

F. substitution

G. linear combinations

H. graphing

3. A snack bar sold 50 items: hot dogs for $3 and drinks for $2. It made $130. How many of each did it sell?

> **Tip:** Write one equation for items and one for money.

Items: h + d = _____

Money: 3h + 2d = _____

h = _____, d = _____